

effects of storing elements with long half lives

economic:

- high cost of storage
- reduction in tourism
- loss of farming produce

social:

- fear of cancer
- sickness in people/animals
- local objections
- people move away

environmental:

- crop mutations
- leakage into water supplies
- pollution of air

ionising radiation and their nature

alpha:

helium nucleus

2+ charge

2n 2p

beta:

high energy, fast moving electrons

-1 charge

gamma:

electromagnetic radiation

high frequency, high energy

ionising radiation and what stops them

alpha:

5cm of air

thick paper

beta:

any named metal/glass/concrete

1m of air

gamma:

20cm or more of lead/thick lead

50cm or more concrete

what is meant by ionisation of a gas

electrons gained or removed from the atom
ions formed

why is alpha more ionising than beta

larger charge
slower moving
more massive
greater volume
more energy